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Irregular Figures

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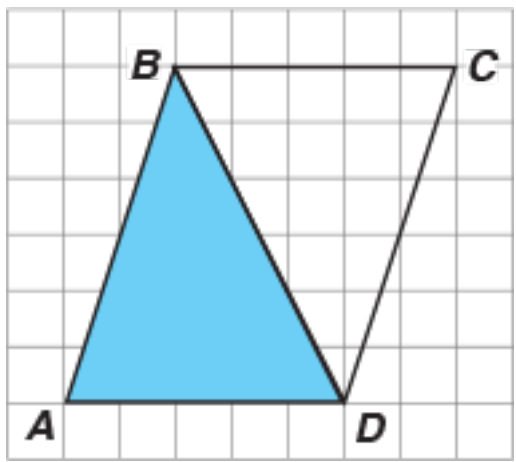
Notes

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Highlights

Irregular Figures

Triangles are related to parallelograms. Study the drawing below. Compare triangle ABD to parallelogram $ABCD$. What do you notice about the area of the shapes? How are they related?



$$A = \frac{1}{2}bh$$

The area of the triangle is one-half the area of the parallelogram. If you know one, you can solve for the other. Parallelogram $ABCD$ has a base of 5 and a height of 6. Find the areas of both figures.

Area of $ABCD$: base $\times$ height
 $5 \times 6 = 30$ square units

Area of ABD : base $\times$ height $\div 2$
 $5 \times 6 \div 2 = 15$ square units

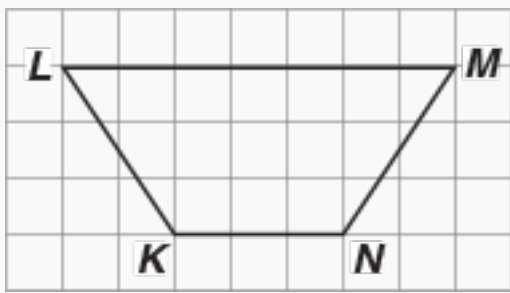
You can write the following formula based on the example.

Area (A) of a Triangle: $A = \frac{1}{2}bh$, where b = base and h = height

Example 1

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Look at trapezoid $KLMN$. To find its area, you need to multiply base by height, but a trapezoid has two bases.



The first step in finding the area of a trapezoid is to find the average of the bases.

$$\frac{LM+KN}{2} = \frac{7+3}{2} = \frac{10}{2} = 5 \text{ units}$$

Multiply the average of the bases by the height of 3.
 $5 \times 3 = 15$ square units

You are given the formula for finding the area of a trapezoid on the GED math formula sheet.

Area (A) of a Trapezoid: $A = \frac{1}{2}h(b_1 + b_2)$, where h = height and b_1 and b_2 are the bases

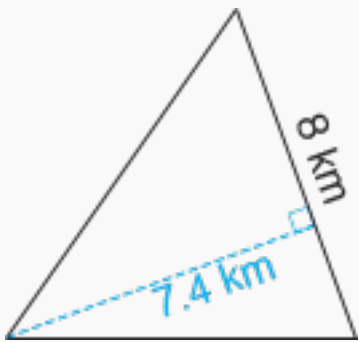
Make sure you understand the formula. Multiplying by $\frac{1}{2}$ is the same as dividing by 2. The formula adds the bases and multiplies by $\frac{1}{2}$, which gives the average of the bases. Then it multiplies by the height. The formula puts the steps in a different order, but it accomplishes the same thing.

As you solve problems, remember that it doesn't matter how the figure is turned. In a triangle, any side can be a base. In a trapezoid, the parallel sides are bases. For all figures, the height must be at a right angle to the base.

Example 2

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In this figure, the base is 8 and the height is 7.4.



$$A = \frac{1}{2} \times 8 \times 7.4 = 29.6 \text{ km}^2$$

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Lesson 2: Perimeter and Area

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